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# Governing Irrationality, or a More Than Rational Government?

# Reflections on the Re-Scientisation of Decision-Making in British Public Policy.

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## Abstract

It appears that recent debates within human geography, and the broader social sciences, concerning the more-than-rational constitution of human decision-making are now being paralleled by changes in the ways in which public policy makers are conceiving of and addressing human behaviour. This paper focuses on the rise of so-called Behaviour Change policies in public policy in the UK. Behaviour Change policies draw on the behavioural insights being developed within the neurosciences, behavioural economics and psychology. These new behavioural theories suggest not only that human decision-making relies on a previously overlooked irrational component, but that the irrationality of decision-making is sufficiently consistent to enable effective public policy intervention into the varied times and spaces that surround human decisions. This paper charts the emergence of Behaviour Change policies within a range of British public policy sectors, and the political and scientific antecedents of such policies. Ultimately, the paper develops a geographically informed, ethical critique of the contemporary Behaviour Change regime that is emerging in the UK. Drawing on thirty in-depth interviews with leading policy executives, and case studies that reflect the application of Behaviour Change policies on the design and constitution of British streets, the analysis claims that current strategies are predicated on a partial reading of new behavioural theories. We argue that this partial reading of human cognition is leading to the construction of public policies that seek to arbitrarily decouple the rational and emotional components of human decision-making with deleterious social and political consequences.

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## Introduction: Laying the Foundations of the Psychological State.

A significant, but for many people imperceptible, force has been quietly reshaping the operational logics of British public policy since the turn of millennium. This largely intellectual force has centred on nothing less than the nature of the human subject, the relationship between our conscious and sub-conscious selves, and the complex interface between the rational and irrational (Thaler and Sunstein, 2008; O'Leary, 2008). In this paper we frame this complex force under the broad moniker of Behaviour Change policies. Although the notion of Behaviour Change delimits a very broad field of political activity and academic inquiry, we take it to mean something more specific than the simple modification of an individual's actions. Consequently, although actions devoted to promoting behaviour change through means of violence, coercion, education, or financial incentives, are as old as organized political society itself, policies for Behaviour Change are altogether more recent and particular in form than their historical predecessors. Crucially, in the context of the focus of this special issue, contemporary Behaviour Change policies marshal a new set of scientific and quasi-scientific understandings of the bases for human decision-making. Three key insights broadly unite these emerging analytics of decision-making: (1) that contra to prevailing neo-liberal orthodoxies, human decision-making does not predominantly follow a "rational" model of contemplative, economically efficient, and self-serving calculation, but is structurally marked by a tendency towards a limited, or "bounded", form of rationality, which could be characterised as irrational (Simon, 1955; 1982;

Shiller, 2005[2000]); (2) that the irrational[1] nature of decision-making is not simply a character of the human condition, but is a product of the design of the everyday environments that we all inhabit (Ariely, 2008; Norman 1988; 2007); (3) that there is a patterning (or predictability) to irrational decision-making, which enables these decisions to be efficiently governed and militated against (Ariely, 2008).

The aim of this paper is to develop a critically informed analysis of the impacts of such visions of human behaviour on different aspects of public policy. These novel paradigms have largely emerged from the interrelated research of behavioural economists, neuroscientists and behavioural psychologists. The insights of this interdisciplinary milieu have both moral and psychological valence for public policy-makers. At a moral level, related research suggests that there may be good grounds for a more interventionist public sector, which can help to compensate for the behavioural flaws of society. At a psychological level, it appears that there are efficient ways of shaping the environments (broadly defined) within which individuals make their decisions, which can produce more favourable behavioural outcomes without compromising the freedom of choice. It is in the context of these moral and psychological opportunities that Behaviour Change policies have found the necessary conditions for their rise to prominence in a number of western states, including the UK.

We begin with an account of the historical rise of Behaviour Change policies in the UK. While this section reflects upon specific policy initiatives, attention is specifically drawn to the changing scientific understandings of human decision-

[1] We actually prefer the term more-than-rational and explain the utility of this concept later in this paper.

making that have informed their development. Drawing on thirty in-depth interviews with key policy executives, the second section explores some of the political and ethical issues that arise from the more-than-rational framing of human decision-making that is encoded in policies for Behaviour Change. The final section provides a critical analysis of actually existing Behaviour Change policies that have been applied to the form and content of British streets. This section is divided into two case studies: the first exploring the connection between street planning and gambling, the second considering the role of street design on the behaviours of car drivers. We utilize these case studies to argue for the development of a more progressive Behaviour Change agenda in the UK, which rather than using new insights into the nature of human decision-making as an executive tool of government, considers their potential for developing a more-than-rational policy-making process, which embraces the creative potential of inexpert knowledge.

## Rationalizing the Irrational: Tracing the Origins of Behaviour Change Policies in the UK

### MINDSPACE and the Genealogy of Behaviour change Policies in the UK.

On the 2nd March 2010 the Cabinet Office Strategy Unit of the UK government, in association with the Institute for Government, published MINDSPACE: Influencing Behaviour through Public Policy (Dolan et al, 2010). MINDSPACE is the latest in a long list of strategic policy documents produced by the UK government that have explored the potential utility of a new form of behavioural theory for the public sector (see Halpern et al, 2004; DCMS, 2001; DEFRA 2007; DWP, 2006a; DWP, 2006b;

Knott et al., 2008). As the culmination of this intensive time of policy deliberation and experimentation, MINDSPACE provides an insight into the nature of the Behaviour Change rationalities of government.

In the Foreword of MINDPSACE, Gus O'Donnell (Cabinet Secretary and Head of the UK's Home Civil Service) and Sir Michael Bichard (Executive Director of the Institute for Government) establish the political and scientific contexts within which the strategy has emerged (see Dolan et al 2010, iii). Behaviour change policies are the product of a recognition within governmental bureaucracies of the limited efficacy of traditional vectors of state action. Political theorists have, of course, known for some time of the inherent weaknesses associated with compulsive forms of state power such as law and regulation (see Allen, 2003). What is, however, novel about contemporary strategies of Behaviour Change is their appreciation of the flawed economic logics that inform the governmental deployment of taxation and related financial incentives as mechanisms to control human behaviour. Related insights into the so-called "irrationality" of economic decisions have emerged from research in the interrelated fields of behavioural economics, Judgment and Decision-Making theory (JDM), behavioural psychology and neuroeconomics (see Ariely, 2008; Damasio, 1995; Le Doux, 1996; Simon, 1955; 1982; Thaler and Sunstein, 2008). Through the careful study of various forms of actually existing economic behaviour (and contrived experimentally-induced behavioural responses) this interdisciplinary field of research has established an increasingly powerful consensual view on the nature of economic decision-making. At the heart of this behavioural consensus are four key observations that: (1) humans neither choose (because of time constraints), or have the ability (because of lack of readily accessible information), to analyze all of the economic

options that are available to them, and compute which of them is in their best financial interests to pursue; (2) contra to conventional theories of market-action (particularly those following Adam Smith and Chicago School economics), that the supposedly punitive actions that markets take on ill-conceived decisions are not necessarily enough to correct related patterns of decision-making in the future (see here Ariely, 2008); (3) that the physical and procedural environments we collectively inhabit actively contribute to the irrational decision-making humans routinely practise (see Norman 1988); and (4) that when analyzed at a significant enough scale, there is a systematic patterning to the repeated mistakes that we make, which suggests that humans are predictably irrational (Ariely, 2008). It almost goes without saying, that in light of these insights in to the nature of economic decision-making it is difficult for policy-makers to be sure precisely what types of Behaviour Change that taxation measures or financial incentives/penalties will actually lead to (Levitt and Dubner 2005: 15-17).

Although it is important to recognize the distinctive features of the Behaviour Change agenda's conception of decision-making, it is necessary not to draw too strong a line of demarcation between these and prevailing neo-liberal paradigms. In his discussion of the behavioural philosophies associated with the American school of neo-liberalism, for example, Foucault (drawing on the work of Becker, 1962) claims that neo-liberal economics has always sought to encompass the rational and "non-rational" (Foucault 2008: 267-271). The key objective of neo-liberalism is then, according to Foucault, not about facilitating a fully rational decision-making process, but uncovering the ways in which people 'react to reality in non-random ways' (regardless of whether such actions are deemed to be rational or non-rational) (Ibid:

269). According to Foucault, it is a belief in the predictability of both rational and irrational forms of human behaviour that distinguishes neo-liberalism from classical liberalism (and justifies a much greater role for government in guiding personal conduct). Perhaps the Behaviour Change agenda is best conceived of then as a product of the new found ability (particularly following developments in behavioural psychology and neuroscience) to be able to recognize, and act upon, the consistencies that exist in rational and non-rational human decision-making: a fulfillment, not negation, of the ultimate neo-liberal goal of the economization of existence.

As a late neo-liberal strategy, MINDSPACE weaves emerging behavioural insights together to form a practical guide to implementing Behaviour Change that is not only applicable to financial decision-making, but also to issues of personal health, environmental management, and community life. However, in order to understand the broader intent that informs MINDSPACE, it is necessary to move beyond the scientific discourses through which it is justified and to explore its political antecedents. To put things another way, Behaviour Change policies have not simply become an object of public policy deliberation in the UK on the basis of their scientific credentials alone. The conditions necessary for the spread of Behaviour Change orthodoxies are as much political as scientific. In this context, it is no coincidence that MINDSPACE is a product of the UK government's Cabinet Office. The Cabinet Office acts as a form of coordinating hub of the UK state, with an overarching mandate of working to ensure the efficacy and effective integration of government action across the full range of policy sectors. It was during the early years of the New Labour administration that a series of prominent civil servants and policy executives started to embrace the ideas of behavioural economics and psychology. In

a series of interviews we have conducted with these civil servants and policy executives it became apparent that they located their penchant for the newly emerging sciences of behaviour within the broader attempts that were being made in the mid 1990s to re-think the role and function of the state within the embryonic New Labour movement. One particular project appeared important in this context.

In 1995 Demos, an independent centre-left UK think-tank, brought together a group of policy theorists and practitioners to produce the influential Missionary Government report (Perri 6 et al., 1995). This report proposed a challenge to the New Public Management approach to government that was in vogue at the time (with its particular emphasis on the use of market principles such as pricing, tariffs, and enhanced efficiencies as mechanisms to guide governmental modes of operation (see Rose, 1999)), and to consider new ways of pursuing social change. According to the editors of Missionary Government, the key weakness of New Public Management was that it had '[f]ailed to understand the complexity and range of human motivations – which financial incentives alone cannot address' (Perri 6 et al., 1995: v). The restructuring of government, it was argued, would essentially be built upon a more complex understanding of the multiple socio-cultural and economic registers of human decision-making. Although the Missionary Government report did not draw extensively on the insights of behavioural economics or neuroeconomics, in emphasizing the importance of understanding more-than-economic human decision-making as a priority of government it created an intellectual niche that could be occupied by these emerging behavioural sciences. As some of the authors of Missionary Government started to work in and around the New Labour Government in late 1990s and early 2000s (and in the Cabinet Office in particular), it is no surprise

that they should have turned to these new behavioural theories in order to pursue their missionary zeal.

The broader significance of the Missionary Government report is not, however, just that it paved the route for new behavioural sciences in to the heart of government. The Missionary Government report also suggested that recognition of the complex, and cross-sectoral, drivers of human decision-making justified a broader role for a more interventionist government. Perri 6 consequently argued that:

> 'From the perspective of the twenty first century, the economistic view of government's powers in the twentieth century may come to seem as obsolete as the military and imperial view of earlier centuries. We may come to see the role of government in far broader ways' (1995: 2).

Essentially Demos coupled a more-than-rational (and more-than-economic) view of human decision-making, with a political project that would support an expanded role for the state within society. It is, of course, not difficult to see how this argument could be made: if economic stimulation is not a particularly reliable, or predictable, means of changing human behaviour, perhaps governments need to find legitimate ways of intervening within the cultural spheres of everyday life. It is important here to note, however, that the new sciences of human decision-making do not automatically translate into the justification for a bigger, and more interventionist form of state. The mixing of behavioural science and governmental politics appears to depend on the prevailing desires and contingent circumstances of the time.

It is instructive, in this context, to note the changing way in which the current Coalition government in the UK is now mobilizing Behaviour Change policies. While the New Labour administration clearly saw the behaviour change agenda as a way of promoting a less coercive form of state intervention, it ultimately provided a series of techniques that enabled government authorities (in classic Third Way terms) to intervene in a broader range of social welfare arenas. But if the ideas of Behaviour Change enabled New Labour to imagine a less intense form, but broader range, of governmental interventions, it is clear that the Coalition government have seized the agenda as a basis for constructing a much smaller state. In the context of the current fiscal crisis facing the UK state, the Coalition government is systematically deploying Behaviour Change policies as a justifying logic to the cutting of state spending[2]. The most dramatic change, in this context, has been witnessed in the Coalition government's attempts to develop health-based behaviour change partnerships with private companies (including pharmaceutical corporations and the food and drinks industry) and third sector groups (Anon, 2010; Cabinet Office, 2010). This fusing of behaviour change technique and Big Society ideology is giving rise to a form of government that appears to care more about the behavioural traits of the public, but takes less responsibility for the delivery of public policy.[3]

[2] Much of the Coalition Government's behavour change policy is being developed in association with the newly appointed Behavioural Insights Team in the Cabinet Office. This team is enabling a much more systematic deployment of Behaviour Change policies than was witnessed under the New Labour administration. Under New Labour such policies were deployed in a far more ad hoc and experimental way.

[3] It is interesting to note the forms that the Behaviour Change agenda has taken within different national contexts. In the USA, for example, it is clear that Behaviour Change policies (and libertarian paternalism more specifically) has been promoted as a way of conceiving of a heighten role for the state in society following from the Neo-Conservative assault on governmental capacities over the last decade. In France it appears that theories of behaviour change are neither associated with a greater or lesser role for the state within public policy delivery. Instead, through the establishment of a centre for neuroscience in the government's centre for Strategic analysis, it appears that the French state is attempting to use brain science as a way of 'fine-tuning' the delivery of public policy (see Centre d'analyse stratégique, 2010).

This paper does not explicitly focus on what the emergence of Behaviour Change policies reveals about the relationship between the state and changing forms of economic theory and expertise (for a broader discussion of these theme see Cairncross 1996; Hall 1989). Nevertheless it is important to emphasize that the behaviour change agenda involves the transfer of both practical forms of technical knowledge (pertaining to how to most effectively and efficiently shape behaviour) and more strategic insights into the limitations associated within rationalized state expertise. To these ends, the psychological and neuro-biological sciences that under-gird behaviour change policies suggest both how to regulate the emotional registers of decision-making, but also the importance of more intuitive forms of human behaviour (forms of behaviour that by definition cannot be anticipated and managed). As becomes apparent as we move through this paper, the empowerment of a new breed of behavioural experts and policy entrepreneurs in government has meant that the regulatory potential of behaviour change policies has tended eclipsed its more radical potentials.

### *Government after Descartes' Error: Rethinking the Times and Spaces of* Decision-Making

Having explored the political backdrop to the rise of Behaviour Change policies in the UK, it is important to consider precisely how related policies are re-framing the human decision-making process. At a broad, and fairly intuitive level, a former member of the UK Cabinet Office Strategy Unit described this new public policy approach to decision-making in the following terms:

> ‘The metaphor I sometimes use is you know, it’s a bit like post-war clothes for women were really ill-fitting but functional, and the difference between those for us is a really well-tailored suit, you see, they’re both clothes and they sort of do the job, and behavioural economics sort of does that compared to classical economics, in relation to policy design. It moves you from rather rough, crude, stick-man type clothing, into one which is really fitted, to change the metaphor, which goes with the grain of human cognition’ (Former Cabinet Office Strategy Unit Member, interview 2009).

The idea that Behaviour Change policies work with the grain of decision-making cognition is an important one. Working with the grain of human cognition essentially implies recognising the complex socio-biological factors that affect decisions and action. At the heart of this comprehension of human decisions lies the assertion that decisions are composed of a mix of rational/calculative and emotional/intuitive responses to choice situations (see here Damasio, 1995).[4]

An appreciation of the coupling of the rational and emotional registers of decision-making is a product of two scientific processes: (1) observations of actual human decision-making (particularly in behavioural psychology and behavioural economics) (Akerlof and Shiller, 2009; Ariely, 2008; Kahneman et al 1982); and (2) studies of the neural activities associated with the taking of decisions (within neurological science) (Damasio, 1995; Ariely, 2008). Two key insights have emerged from such work.

[4] It is at this point salient to acknowledge that an appreciation of the complex connections that exist between the rational and affective registers of the human condition has provided a basis for coercive as well as more progressive programmes of behaviouralist state programmes. The utilization of shock and trauma as state technologies have, for example, been used at both an individual (interrogation/torture) and collective level (warfare/ambient conflict) as emotional strategies for re-rationalizing human conduct in politically docile directions (see Anderson, 2010; Klein 2008: 3-71).

First, is an increasingly sophisticated understanding of the role of emotions (variously described as 'rapid cognition responses', 'more-than-rational' motivations; and, of course, 'gut reactions') in the making of decisions. Second, is an ongoing concern with the various couplings and loops that connect the emotional and the rational within human behaviour. This second point is where the complexity of human decision-making starts to become apparent (the implications of such developments for geography have, of course, already been extensively explored as part of the "affective turn," see Anderson and Harrison, 2010; Anderson, 2009; Thirft, 2000. See also Barnett, 2008; and Papoulias and Callard, 2010 for a critique of such work).

Certain prominent strands of research in behavioural economics, psychology and neuroscience do not link the emotional and the rational arms of human decision-making in simple, zero-sum terms (unconscious or conscious; automatic or deliberative; error-prone or error-free) but, instead, attempt to unpack the diverse roles of emotions in practices of reasoning, and the impact of reasoning on the formation of emotion (see Damasio, 1995). To these ends emotion and reasoning are no longer seen to be separate practices, conducted in separate neurological segments of the brain, but are conceived as deeply interconnected modalities of the decision-making process. This insight is significant not only because it appears to offer a more accurate (at least neurologically) account of the nature of decision-making, but also because it works against the pejorative characterization of emotions as being an inferior basis for decision-making. Damasio asserts this more nuanced account of the role of emotions in human behaviour when he reflects that '[T]he obligate participation of emotion in the reasoning process can be advantageous or nefarious

depending both on the circumstances of the decision and the past history of the decider' (1995; see also Gladwell, 2005).

Behaviour Change policies in the UK tend to depict the role of emotions in human decision-making in a far less ambiguous way than is evident in contemporary neurosciences. Consequently, while embracing a more emotionally sensitive account of the human decision-making process, contemporary policies for Behaviour Change in the UK trend towards a position where emotions are seen to be playing an overly significant, and thus pernicious, role in everyday human behaviour. This line of argument is inspired predominantly by behavioural economists, and is predicated on a relational understanding of the decision-making process. Behavioural economists argue that decision-making is rarely made in absolute, isolated contexts: it is normally positioned in relation to our surrounding socio-cultural and physical environment[5] (see here Ariely, 2008). They refer to the contingent environments within which we make our everyday decisions as choice architectures (see Thaler and Sunstein, 2008, 81-100). Crucially, Behaviour Change policy is based upon the assumption that the choice architectures we routinely inhabit are biased in such a way that they actually make it more difficult for us to make the decisions that would be in our own best

[5] It is important to note here that the figure of the 'environment' has become a prime object of intervention associated with the British behaviour change agenda. It is also not incidental that in British policy circles the notion of choice architecture has been gradually replaced by discussion of choice environments. The notion of environment is utilised to deliberately incorporate a range of contextual factors (physical setting, psychological framing, visible landscape, and social network) that extend beyond architectural concerns. The idea of the choice environment appears to have emerged out of interrelated work within cognitive design (see Norman, 1988), criminology (particularly the 'broken-windows theory of crime), and broader sociological analyses of the 'power of context' (see Gladwell, 2000: 133-168). The focus on environmental drivers of behaviours is significant because it tends to dislocate the decision-making process from the human subject and into the realms of public space and legitimate governmental intervention. The emphasis placed on environmental targets also tends to axiomatically emphasize the more irrational/emotional components of decision-making, as these are the decision-making processes that appear to be most susceptible to the affective pushes of environmental context.

interests. Consequently whether it is our failure to invest sufficiently within a pension scheme, eat healthy food, take regular exercise, manage our personal finances, or conserve domestic energy use, behavioural economists would see such shortcomings as products of the prevailing norms that our social and physical environments tend to produce (see Halpern et al., 2004). In other words, the social and commercial environments that we inhabit are seen to exploit the emotional dimensions of our decision-making apparatus to produce human behaviour that tends to be characterized by pernicious short-termism.

The phrase that is often used to capture how choice architectures generate the routine reproduction of bad decisions is arbitrary coherence (see Ariely, 2008). The notion of arbitrary coherence conveys how seemingly natural choices about our consumption patterns and welfare are in fact highly random outcomes of our choice environments, that only take on coherence (or the veneer of logical consistency) because they become an habitual part of our personal and collective cultures (see Knott et al, 2008). What is interesting about notions of arbitrary coherence, in relation to the themes being explored within this special issue, is that it not only questions the spaces within which we make decisions, but also the timeframes of our behaviour. A key dimension of theories of arbitrary coherence is the realization that our decisions are not only conditioned in relation to the environments within which they are made, but also in relation to the previous decisions we have made in the past (ibid). Behavioural economists have thus observed that while our decisions are anchored by the environments in which we make them, they are also structured by the nature of the first decisions that we make in relation to a variety of issues. This process of what is termed imprinting reflects the fact that, despite its often arbitrary nature, the first

decision we make on a range of issues (to drive rather than walk to work in the morning; to deposit our money in one savings account as opposed to another; to take breakfast at the fast food joint rather than the organic food shop) tends to impact on a long succession of subsequent decisions (ibid). It is because of the observed impact of imprinting on human behaviour that Behaviour Change policies not only attempt to extend the decision-making moment in to the future (in order to encourage better long-term planning), but also in to the past (in order to understand why certain forms of potentially harmful behaviour have become de rigour).

In the context of perceiving decision-making moments as the relational outcome of arbitrary environmental and historical choice architectures, Behaviour Change policies seek to actively restructure the spatial and temporal situations that frame decisions. In the UK, this restructuring process has, at one level, attempted to facilitate the more rational components of the human decision-making process. This re-rationalization process involves the use of social marketing, the mobilization of respected agents of persuasion (such as doctors, sports personalities, and community leaders), and the production of more accessible forms of information, in order to enable individuals to think about the actual impacts of their habitual behaviours and break the cycles associated with arbitrary cohesion (see COI, 2009; Halpern et al 2004; Knott et al, 2008). At other times, UK policies for Behaviour change have exploited insights in the emotive/automatic nature of human decision-making as a basis for restructuring choice architectures in ways that encourage new forms of action without necessarily requiring individuals to think about the changes they are making. Such policies can involve the setting of favourable default positions (on

pensions schemes and organ donation registers[6]) or the more careful planning of buildings and community spaces so that it is easier to make healthy choices (see Department of Health, 2004; Department for Transport, 2007; Department for Work and Pensions, 2006b). In recognition of the impacts of behavioural imprinting, a strong emphasis has been placed within UK policies for Behaviour Change on the importance of instigating change at salient times in peoples' lives, and of carefully staging behavioural interventions at appropriately early points within the decision-making cycle (see Dolan, 2010; Halpern et al, 2004).

It appears that contemporary public policy in the UK is marked by a series of quite specific interpretations of the nature of the connections that exist between the rational and more-than-rational components of decision-making. In many ways these interpretations appear to be driven more by the strategic requirements of public policy delivery than broader ideological forces. In the following section we highlight some of the ethical issues that such forms of Behaviour Change policies raise as we begin to develop a more critical theory of the behavioural turn in public policy.

**Liberal Orthodoxies and the Ethical Challenges of Behaviour change Policies**.

The previous section emphasized how the rising prominence of new theories of human behaviour within British public policy have led to a heightened concern with the emotional component of the decision-making process. The scientific analyses that have gradually revealed the inherent emotional dimensions of human decision-making

[6] We note here that controversies over the use of presumed consent within British organ donation has so far prevented the use of such default settings within the UK's Organ Donor Register.

have led to increasingly complex understandings of the nature of human behaviour that suggests that the rational and more-than-rational drivers of action are actually different parts of the same cognitive system. It appears, however, at least in the context of the UK, that the interpretation of human behaviour as emotionally constituted has been coupled with a normative assessment of the more-than-rational, automatic functions of the brain as being pernicious influences on decision-making. This normative perspective is evidenced in the fact that Behaviour Change policies have either sought to create choice architectures that perpetually seek to stimulate the rational faculties of decision-making, or use the emotional triggers of behaviour as a behavioural vector through which to stimulate rational action. We claim that these normative approaches to the emotional aspects of decision-making raise significant ethical challenges for Behaviour Change policies as they are currently being conceived.

The first set of ethical concerns relates to the ways in which the emotional and rational dimensions of decision-making are used to mark out distinct socio-behavioural categories within the general population. This ethical issue was distilled in an interview we conducted with a public policy expert at a prominent right of centre think tank when they observed,

> ‘For people who are relatively intelligent and sophisticated it doesn’t really matter because if you’re being nudged and you understand you’re being nudged and you know what’s in your best interests and all the rest of it then you can

> work round the nudge fairly easily […] One worry that a libertarian paternalist[7] citizen would be somebody who, when is nudged into certain types of behaviour, which proves then to be not in their best interest, will be somebody who'll want to take some kind of action against the government for compensation' (Institute of Economic Affairs Interview, 2009).

Two issues stand out in this statement. First of all is the apparent decoupling of the emotional and rational components of the human decision-making process and their redistribution onto different social groups who are associated with more or less rational tendencies. The suggestion that Behaviour Change policies are designed with certain target populations in mind is indicated by official government support for socially segmented delivery systems for such policies (see DEFRA, 2007)[8]. These types of policy formulation serve to construct an artificial separation of the rational and irrational, or at least to designate discreet social groupings that are more and less able to flex their rational capacities against their emotional impulses (to place this process in longer historical context see Valverde, 1998). Not only is this a normative misreading of emerging understandings of the irrational constitution of decision-making, but it also runs the very real risk of constructing stigmatized social classes. This decision-making underclass is already being marked out geographically as well as demographically by the operation of such policies (Thompson et al, 2007).

[7] Libertarian paternalism is a particular approach to Behaviour Change policies that emphasizes the importance of preserving elements of choice while constructing paternalist choice architectures (see Thaler and Suntein, 2008).

[8] The Department of Environment, Food and Rural Affairs operates a social segmentation model for the delivery of their policies. This segmentation model breaks down the population into seven separate groups (ranging from "stalled starters" to "positive greens"), which are associated with bespoke policy strategies for achieving Behaviour Change (see DEFRA, 2007).

The second issue raised within this quote is the role of the state within the Behaviour Change process. The concern with litigious retribution expressed here serves to highlight two interrelated ethical issues. The first concerns exactly who, within bureaucratic networks of government, will be responsible for setting the rational default options and logical choice architectures that will guide flawed human behaviours. To address this point requires more than revisiting the traditional Weberian discussions of the constitutional legitimacy of the expert and technocrat within the democratic state (Weber, 1968). It concerns a deeper ontological question: namely the extent to which, in light of the new sciences of human behaviour, state officials can claim the forms of rational distance that have historically been associated with government systems (see Mulgan, 2009). We assert that in this formative period of Behaviour Change policy in the UK, the state is at best a dilettante; wielding a largely unfounded belief in the predictability concerning the nature of human decision-making.

Our second, and perhaps more obvious set of ethical concerns, pertains to the deployment of often sub-conscious forms of psychological power within a liberal democratic state system. As has been previously mentioned, a central tenet of contemporary Behaviour Change policies in the UK has been a desire to utilise the new forms of scientific understanding of the automatic, and often subconscious, drivers of decision-making as a basis for developing suitable prompts for desirable action. The forms of state intervention in to the collective subconscious that have been built on these new behavioural insights raise significant questions concerning the legitimacy of governmental involvement in the unconscious realms of everyday life.

The democratic dilemmas surrounding Behaviour Change policy were captured in one interview we convened with a prominent policy executive,

> 'Now, the best argument in Nudge remains that choice architecture is not neutral […] you have to twin it to collective responsibility, so you have to get the permission of citizens to use the techniques, rather than just banning mars bars from by checkouts, you have to say, "what are we going to do about mars bars by checkouts?" should it be fresh fruit, or something, so you're kids don't nag you because you can't do that just at the individual level […] you have to collectively reach a view, so that means you need some kind of collective mechanisms, by which you give government or its agencies, permission, as it were, to do the manipulation' (Former Cabinet Office Strategy Unit member, interview 2009).

The reference made here to Thaler and Sunstein's influential Behaviour Change manual Nudge: Improving Decisions About Health, Wealth and Happiness (2008) is an important one. In Nudge Thaler and Sunstein tackle head on what they describe as one of the main ethical challenges to Behaviour change policies: namely that they act in insidious ways and enable, often unelected, government officials to manipulate human behaviour in directions that best suit those in power (2008: 244).

Such concerns rest on the connection between decision-making and the constitution of liberal democratic society. The ability to make unencumbered decisions, so long as these decisions do not lead to harm being caused others, has been an enduring liberal orthodoxy (Mill 2002 [1859]). This historical mode of limitation on the power of the

state is clearly challenged by Behaviour Change policies. First, it is clear that current Behaviour Change regimes seek to extend the legitimate reach of governmental action from the realm of causing harm to others to instances of inflicting harm on the self. Second, and more critically, when related policies exploit the emotive, automatic drivers of decision-making through subconscious priming or default settings, there is a very real danger that the decision-making process (at least in its full neurological richness) is by-passed, and its role in upholding liberal norms eroded.

Popular concerns over Behaviour Change policies routinely emanate, moreover, from a realization that many of the mechanisms utilized in such policy regimes are borrowed from the corporate world, where smart marketing, product placement, and subtle forms of psychological manipulation has become something of an art form (see Frank, 1997). The misapplication of such forms of power has been exposed in Connolly's analysis of the affective intensities that were brought to bear on the electorate by the right-wing political-media complex during the 2004 Presidential election campaign (Connolly, 2005: 879-880). While the psychological shenanigans of election campaigns is perhaps to be expected, Behaviour Change policies are suggestive of a much more prosaic misapplication of neurological power within the details of our everyday lives. It is in the context of such emerging concerns that certain Behaviour change policies have been under-girded by a series of libertarian principles of action and limitation. As is intimated in the previous interview quote, a series of policy experiments have already been conducted in the UK to explore potential democratic mechanisms for the implementation of Behaviour Change policies. While acknowledging that government cannot address the public's potential desire for the "right to be wrong", MINDSPACE described the importance of

deploying “democratic engagements” and of gaining public permission in support of Behaviour Change regimes (Dolan at al 2010: 63-72). Thaler and Sunstein (2008: 243-246) have been, perhaps, less concerned with democratic processes when constructing their principles of limitation. They argue that the public disclosure of Behaviour change policies, and the transparency of associated mechanisms, is enough to ensure their libertarian legitimacy. As we will explore in greater depth in the following section, our main concern with such principles of Behaviour Change limitation is that they tend to exploit a rather narrow set of democratic procedures to justify pre-determined desired policy outcomes. This not only tends to reduce democracy to the taking of pre-selected choices, but fails to recognize the potentially radical impacts that new behavioural theories could have on how we perceive the constitution of the democratic decision-making process itself.

**The View from the Street: Planning, Design and the Contested Practices of Behaviour Change.**

In order to explore further the nature, contradictions, and opportunities associated with Behaviour Change policies in the UK it is important to consider specific policies as they operate in practice. In this section we focus on the impact of Behaviour Change policies on the planning and design of the UK’s streets. The form and function of streets have provided an important fulcrum for Behaviour Change policy experimentation and implementation in the UK. The street has become significant in such policy developments because it constitutes one of the most immediate public choice architectures we are routinely confronted with in our everyday lives (Allen, 2006; Barnett, 2008; Huxley, 2006; Minton, 2009; Rose et al 2010). To these ends

we utilize the street as a context within which to critically interrogate the ethical implications of Behaviour Change policies, and to illustrate the value that we believe geographical enquiry can bring to this endeavour.

## Gaming the Street: Regulating Ambient Gambling and Controlling Addictive Behaviours.

In the Haringey political constituency of north London there are currently 72 betting shops that centre on three of the most deprived locations in the area (Pears, 2010). On one 300-yard stretch of the Green Lanes road alone nine betting shops have clustered together (ibid). The blanket saturation of Haringey by the gambling industry has become an issue of significant political contestation in north London, with the local Member of Parliament, David Lammy, campaigning strongly against the location of new betting shops in his constituency, and Ken Livingstone taking up the cause as part of his bid to be re-elected as Major of London (ibid). But while extreme, the processes occurring in Haringey have been replicated to greater or lesser extents on streets throughout the UK (Toynbee, 2008). It was in the twin context of the growing success of the UK's betting industry, and its association with the generation of a range of personal financial problems, that the Home Office Commissioned the Gambling Review Report of 2001. The findings of the Report (which ultimately become the responsibility of the Department for Media, Culture and Sport (DCMS, 2001)) interpreted gambling through understandings of the decision-making process that had been influenced by Behaviour Change philosophies.

At one level the Gambling Review Report connected certain forms of gambling (certainly those connected to the study of predictable patterns) with advanced cognitive skills. It also argued that in most instances gambling was a rational process within which "punters" are fully cognisant of the high probability that they will lose (DCMS, 2001: 3). But in classical behavioural economic parlance, the Report goes on to recognize the ways in which gambling can stimulate the more emotive and less rational registers of human action,

> 'We recognise that some individuals become obsessed by gambling to the point at which they cease to function as normal members of society and may do great harm not only to themselves but also to their families and possibly to the general public. We believe that it is a legitimate role of regulation to limit the risk of problem gambling even if this means restricting the freedom of those who can gamble harmlessly' (ibid: 4)

The reference made here to "some individuals" who actually lose their rational decision-making capacity in the processes of gambling is troubling because of the way, as with so many Behaviour Change policies, it isolates and naturalizes irrationality within certain segments of the population. While the propensity for addictive forms of gambling behaviour may or may not be associated with certain biological traits, it is clear that in the targeting of poor, working class communities (those communities whose relative levels of income make the potential gains to be made through gambling that bit more appealing) the gaming industry actively contributes to the socio-economic production of an addictive class. Notwithstanding this critique, the Report does recognize the potential harm caused by the architectures

of gambling choice that currently characterize many UK high streets. The problematic choice architectures are articulated in the Gambling Review Report as ambient gambling:

> 'We were unwilling to see an increase in ambient gambling, that is, gambling opportunities that are available in locations which are not dedicated to gambling. We also wish to limit the extent to which gambling could be combined with the consumption of alcohol. We do not therefore propose permitting betting in pubs, or alcohol in betting shops […] We propose that gaming machines be banned from premises such as cafés and taxicab offices [and] not be permitted as "exempt entertainments" (ibid page 4).[9]

The government's official response to the Review, A Safe Bet for Success (DCMS, 2002), and the subsequent Gambling Act (2005), largely endorsed its findings and recommendations. In recognizing the purported economic and cultural importance of gambling to UK society, the government sought to develop a set of policies that would allow the development of a globally competitive British gaming industry, but would also protect society from the potentially damaging consequences of this expansion. The insights of Behaviour Change thinking have clearly been influential on these terms, as they suggest ways of facilitating rational forms of financial decision-making while not restricting the ability of individuals to make the decision to gamble. The government attempted to stop the spread of ambient gambling through restrictions on the locations in which gambling could take place and through setting

[9] In addition to these locational methods of gambling regulation, the Gambling Review Report also recommended a mandatory code of practice for social responsibility within the gaming industry. Limits to prizes and payments to/from gaming machines in family entertainment areas were recommended to protect children: the report points out that most other countries in the western world ban children from all gambling (DCMS, 2001 page 4).

minimum floor space requirements for new gambling premises (thus stopping the spread of multiple, small gambling establishments). The government has also developed policies that reflect the forms of cognitive models promoted by advocates of Behaviour Change. Through the enforcement of new codes of practice for licensed gaming locations, the government now requires that businesses enable the irrational tendencies of the gambler to be coupled with technologies of rationalization. Such technologies include spending monitors and reality check facilities, which are specifically designed to counteract the emotional flow of gambling decision-making processes. Perhaps the most significant aspect of the code of gambling practice implemented as part of the 2005 Gambling Act was the self-exclusion option. Self-exclusion programmes are prototypical Behaviour Change policies to the extent that they recognize how the seductive passions of the gambling experience can make participation difficult to resist for those who suffer from addictive tendencies. Self-exclusion thus works by separating out (in both time and space) the rational choice not to gamble, from the less-than-rational impulse that may take hold when confronted by gambling establishments. Once the decision not to gamble has been made, it is expected that gambling establishments will enforce these self-imposed restrictions, with the assistance of the police if necessary (Gambling Commission, 2008 page 32).[10]

Given the clear attempts that have been made to regulate the decision-making environments surrounding gambling in the UK, it is necessary to consider precisely why high streets appear to be becoming increasingly conducive environments for

[10] Using a self-exclusion agreement replete with photographs, the so-called "problem gambler" can exclude themselves from a range of establishments simultaneously and for varying lengths of time. The agreement also enables agencies to legally share information about them – preventing the time-consuming task of self-excluding oneself from individual premises.

ambient gambling. The 2005 Gambling Act in aggregate clearly reflects an attempt by the UK government to chart a strategic course of action in support of both the British gaming industry and those groups concerned with the socially deleterious affects of gambling. In terms of regulation, it thus attempts to stop the spread of so-called super-casinos on the one hand, and smaller-scale forms of ambient gambling on the other. It does, however, clearly seek to appease the economic needs of the gaming industry by allowing the spread of traditional betting shops. The 2005 Gambling Act thus makes provision that licensing authorities should permit the use of premises for gambling unless it is thought that such premises would endanger vulnerable persons or promote crime and disorder, without need for an assessment of the levels of demand for such a development, and provided that assistance is made available to those who could experience problems by being exposed to new gambling opportunities (Gambling Act 2005: Section 24; Section 153). Given the presumed consent surrounding the awarding of licenses to high street betting establishments, it should come as little surprise that Haringey has witnessed a congregation of betting establishments within its constituency. Two things are, however, of particular significance within the 2005 Gaming Act. First, restructuring of the types of choice architectures that promote arbitrary forms of coherence in betting behaviour is not simply a technical planning or licensing issue: it is a political process that requires the strategic management of different social and financial visions of the high street. Second, the example of the 2005 Gambling Act illustrates how Behaviour Change policies in the UK are routinely based on a selective reading of the behavioural science on which they are based. In this instance, for example, it appears that so long as the irrational decision-making of vulnerable individuals (who supposedly cannot consistently activate their rational cognitive functions in a consistent way) can be regulated through the

psychological policy fixes provided by Behaviour Change policies, the spread of gambling deeper into community life in the UK is acceptable. This kind of policy rationale does not seem to appreciate the emotive responses that we all may have to an environment that is teeming with ambient gambling opportunities, or the longer term imprinting such opportunities to gamble may have on our future conduct.

**Sharing space and psychological speed bumps: changing driver behaviour on residential streets.**

The second example of street-based Behaviour Change policies explored within this paper concerns recent attempts to control the behaviours of drivers on the UK's residential roads. Much has, of course, already been written on the impacts of driving on human behaviour and decision-making (see here Merriman, 2007; Pica et al, 2008). It appears that within the isolated environment of the automobile there is a tendency for humans to alter their behaviour in a range of ways. Some research has shown that people tend to become more aggressive and single-minded when driving, which often results in unsafe driving speeds and the taking of greater than normal risks (Department for Transport, 2007). It is in the context of the association between driving and more emotive registers of decision-making that the principles of Behaviour Change have become popular within road safety campaigns and traffic management strategies (Department for Transport, 2007; Department of Environment, Transport and the Regions, 2000).

The joint publication of the Manual for Streets in March 2007 by the Departments of Transport and Communities and Local Government, marked a significant turn towards the principles of Behaviour Change policies in street planning and design in

the UK (see Department for Transport, 2007). The Manual for Streets details design guidance for the construction of new residential streets and for the restructuring of existing streets (for a more detailed discussion of this policy see Anon, forthcoming). The design philosophy at the centre of the Manual for Streets is that residential streets need to be planned in ways that do not only focus on the movement functions of roads, but also on their role as places where people live, work and play (ibid: page 2). The key assumption under-pinning this philosophy is that as a choice architecture residential streets have actively contributed to the harmful behaviours that we associate with drivers. In this context, the Manual for Streets claims that the construction of linear streets of increasing width, which carefully segregate place and movement functions (that is people from motor vehicles) have actively encouraged dangerous driver behaviours.

Drawing on the insights of new behavioural theories and recent developments within Dutch road design, the Manual for Streets suggests a series of innovative ways in which the psychological infrastructures of residential streets can be usefully changed (see Elliott et al 2003; Kennedy et al 2005; Lawton et al 2003). Two key psychological insights frame the design guidance set out in the Manual for Streets. First is the importance of developing psychological prompts that indicate to drivers that the street that they are on is a place where people live and interact. The strategic placement of planters, the building of archways at street entrances, the construction of narrower, non-linear street geometries, and the painting of road surfaces in creative ways are all recommended as effective psychological speed bumps to motorists, who suddenly find that the traditionally de-homogenized spaces of the street has numerous place-based references. The second set of psychological strategies set out in the

Manual for Streets encourages the design of increasingly shared spaces, where traffic and people can interact. By creating shared traffic/pedestrian spaces (through the removal of curb stones, road makings, and unnecessary road signage) it is anticipated that planners can further enhance the place-based feel of the street, while also fostering more intuitive cognitive systems of shared responsibility among drivers and pedestrians. It is argued that by forcing cars and people to interact, both drivers and pedestrians will become far more skilled in reading the body language and intensions of all street users, and develop appropriate strategies to ensure the effective functioning of streets.

What is interesting about the promotion of shared spaces within the Manual for Streets is that it actually encourages the development of more-than-rational, automatic response environments to address the problems of road safety. To these ends, and unlike many strands of Behaviour Change policies in the UK, the Manual for Streets does not construct a normative distinction between the rational and emotional dimensions of human decision-making. It essentially recognizes that solutions to problematic behaviour may actually be found in the utilization of emotional as well as rational bases for decision-making. In the Manual for Streets we can thus discern the deployment of Behaviour Change policies that value the training and marshalling of more and less rational ways of making decisions. This is a policy position that finds support in the increasingly complex picture of human cognition that is currently emerging within neuroscience (see here Damasio, 1995), not to mention longer established feminist philosophies of mind (e.g. Prokhovnik, 1999).

Our provisional research into the practical application of the Manual for Streets, however, reveals an important limitation being placed on the role of more intuitive decision-making in contemporary street design (see Anon, forthcoming). In light of the concerns that the professional ranks of urban planners and traffic engineers have expressed in relation to the potential health and safety implications of shared space street design, the implementation of the *Manual for Street's* design principles has been colonized by the rationalizing practices of the government expert. While this has not prevented the implementation of shared street spaces – and the intuitive systems of behaviour they are supposed to cultivate – it has prevented the utilization of more intuitive design insights from those residents who use and know the street on a daily basis, from being effectively incorporated in the design process. Not only does this failure to embrace more emotional responses to questions of street design reflect a partial application of contemporary Behavioural Change theories, it also tends to work against one of the cognitive grounds upon which local people could claim a more deliberative and democratically inclusive role within the actual design of Behaviour Change policies.

**Conclusion: From Instrumentalism to the Valuing of the Inexpert.**

This paper has charted the rapid development of a psychologically informed, Behaviour Change armature within the UK state. While predicated upon seemingly strong scientific credentials, we have shown how the application of the findings of neuroscientists, behavioural economists and psychologists is subject to significant political influence. The politicization of theories of Behaviour Change can be ideologically driven (particularly when it is connected to justifications for either a more or less interventionist state), or simply the product of the practical requirements

of public policy delivery (seen in relation to the simplification of current understandings of the emotional dimensions of decision-making). But our attempt to expose the highly contingent readings of the nature of human decision-making within contemporary public policy in the UK derives not from a desire to reveal the inevitable corruption of science by politics. It is instead inspired by our wish to suggest that the type of state that emerges from current theories of human behaviour can be directed in a range of creative and inclusive ways.

This paper has illustrated how emerging theories of human decision-making and behaviour are being utilized as tools for the delivery of pre-existing policy goals. The instrumental utilization of such behavioural insights leaves little opportunity to question the political and economic values that may lie behind the policy goals of the Behaviour Change agenda. We argue that the instrumentalization of Behaviour Change policies is a product of the separation of the rational and more-than-rational aspects of decision-making within public policy in the UK. This separation process, and the normative categorization of the emotional divers of behaviour as anathema to human health, wealth and happiness, enables policy-makers (and an emerging cartel of psychocrats[11]) to foreclose discussions of what the values associated with "good behaviour" should be. If humans are seen as slaves of their emotional selves, it appears that only those choice architects who design public policy can be trusted to design the rational default environments in which we are to live (with of course the exception of the "rational elite" for whom these policies are not really meant in the first place). It is this conceit that lies at the root of both the democratic shortcomings

[11] We use the term psychocrats to refer to the diverse range of policy-makers, planners and designers who are currently using the insights of behavioural theories to shape human behaviour without pursuing input from or consultation with the publics who are directly affected by such strategies. To these ends, psychocrats embody a new class of psychologically empowered experts who remain largely unaccountable and unidentifiable actors within the bureaucracies of the state.

of Behaviour Change policies, and its unfortunate psychographic designation of irrational social groupings.

There are two questions we would like to end this paper by asking: 1) what if all humans are bound by the same more-than-rational tendencies when making decisions? And 2) what if this emotional component of human behaviour was seen as a potentially positive driver of human behaviour? The answer to both of these questions has, of course, already been suggested in this paper and within much of the most respected work emerging from neuroscience and behavioural psychology. In answer to the first question, all humans are bound (admittedly to greatly varying extents, and at different times and in different spaces) by the same irrational/emotive/automatic tendencies within our decision-making, and we would all be significantly less effective at making all of the daily decisions we have to make if we could not rely on these integrated cognitive abilities. In answer to the second question, more-than-rational responses to decision-making situations are not only often advantageous, but are also often a necessity. To these ends, the lessons of contemporary behavioural theories for public policy in the UK appear, at present, to be being only partially applied. If the human dependency on more-than-rational responses leads to harmful patterns of behaviour then there is a clear role for a rationalizing government to try and redress these situations. However, an alternative, but nonetheless valid, interpretation of behavioural sciences is that governments should place greater trust in the role of human emotion and intuition in the design and application of policy.[12]

[12] It is important to acknowledge at this point that the positive valuation of the emotional and intuitive components of human decision-making (particularly its association with a very localized and embodied sense of self-determination) is based upon a particular interpretation of the sciences associated with study of human cognition. Papoulias and Callard (2010) have recently been highly critical of the ways

There is potentially much that can be learned from more-than-rational perspectives on the world, as they lead policy makers into a new appreciation of the inexpert perspective and local vantage point. The critical issue here is the value of public policy that challenges situations when either the irrational or rational dimensions of decision-making come to dominate the decision-making process to the detriment of the decision-maker's welfare or freedom.

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in which the "affective turn" and non-representational theorists have utilized the neuro-biological work of scientists like Damasio and LeDoux in order to claim that the intuitive registers of human cognition provide the basis for freer forms of human association with and being in the world. Papoulias and Callard (2010) emphasize that although popular neuroscience emphasizes the important role of the emotional within human decision-making, the emotional content of any decision is not seen as an indicator of freedom (from rational inscription), but a pre-set function of the evolutionary dynamic of species (41). Utilizing the non-rational within public policy development may thus not lead to a pre-cognitive liberation, but rather dependency upon a biological inscribed substratum of our behavioural instincts.

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